

Introduction to Microprocessor and Microcontroller

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Why do we need to learn Microprocessors/Microcontrollers?

- The microprocessor is the core of computer systems.
- It's not an overstatement if it says that, today there is no electronic gadget on the earth which is designed without a Microcontroller. Ex: communication devices, digital entertainment, portable devices etc...
- Nowadays many communication, digital entertainment, portable devices are controlled by them.

Why do we need to learn Microprocessors/Microcontrollers? (cont'd.)

- **Personal information products:** Cell phone, pager, watch, pocket recorder, calculator
- **Laptop components:** mouse, keyboard, modem, fax card, sound card, battery charger
- **Home appliances:** door lock, alarm clock, thermostat, air conditioner, TV remote, VCR, small refrigerator, exercise equipment, washer/dryer, microwave oven
- **Industrial equipment:** Temperature/pressure controllers, Counters, timers, RPM Controllers
- **Toys:** video games, cars, dolls, etc.

Why do we need to learn Microprocessors/Microcontrollers? (cont'd.)

So, A good designer should always know what type of controller he/she is using ,their architecture, advantages , disadvantages , ways to reduce production costs and product reliability etc....

Microprocessor

- Microprocessor and CPU are used interchangeably.
 - A microprocessor is a multipurpose, programmable, clock-driven, register based electronic device that reads binary instructions from a storage device called memory, accepts binary data as input and processes data according to those instructions and provides results as output.
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Microprocessor(cont'd.)

- A CPU built into a single VLSI chip is called a microprocessor.
- It is a general-purpose device and additional external circuitry are added to make it a microcomputer.
- The microprocessor contains arithmetic and logic unit (ALU), control unit, Instruction register, Program counter (PC), clock circuit (internal or external), reset circuit (internal or external) and registers. But the microprocessor has no on chip I/O Ports, Timers, Memory etc.
- For example, Intel 8085 is an 8-bit microprocessor and Intel 8086/8088 a 16-bit microprocessor.

Microprocessor(cont'd.)

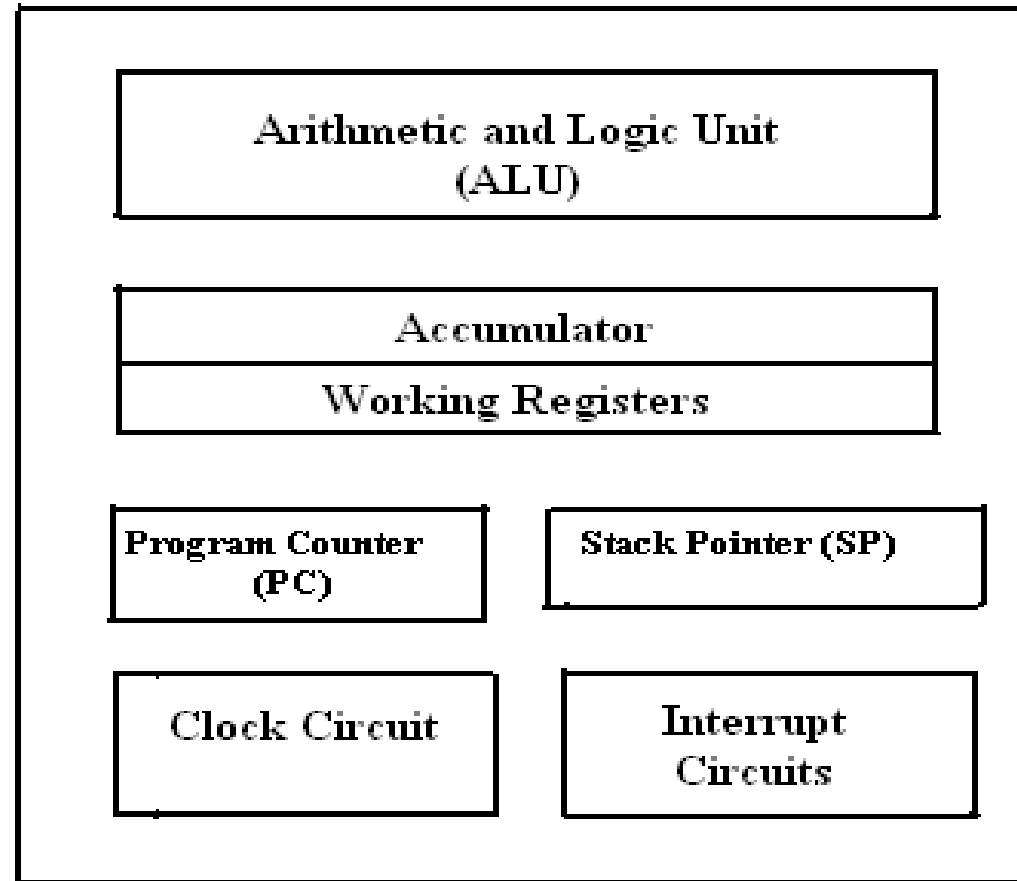


Fig. Block diagram of a Microprocessor.

Microprocessor(cont'd.)

Arithmetic and Logic Unit (ALU)

- It is the place where actual computations take place.
- Consist of circuits which performs arithmetic operations over data received from memory and capable to compare to compare numbers.

Microprocessor(cont'd.)

▶ Control Unit (CU)

- The coordination and control of a computer is the sole responsibility of the CU.
- It interrupts the instructions fetched into the computers, determines what data if any are needed, where it is stored, where to store the result of the operations.
- It ensured that according to the fetched instructions the right operation is done on the right data at the right time.

Microprocessor(cont'd.)

▶ Registers

- Registers are temporary storage locations inside microprocessor that hold data and address.

Microprocessor(cont'd.)

▶ Program counter (PC)

- It is commonly known as IP (Instruction Pointer) in 8086, is a register in microprocessor that contains the address of the next instruction to be executed.

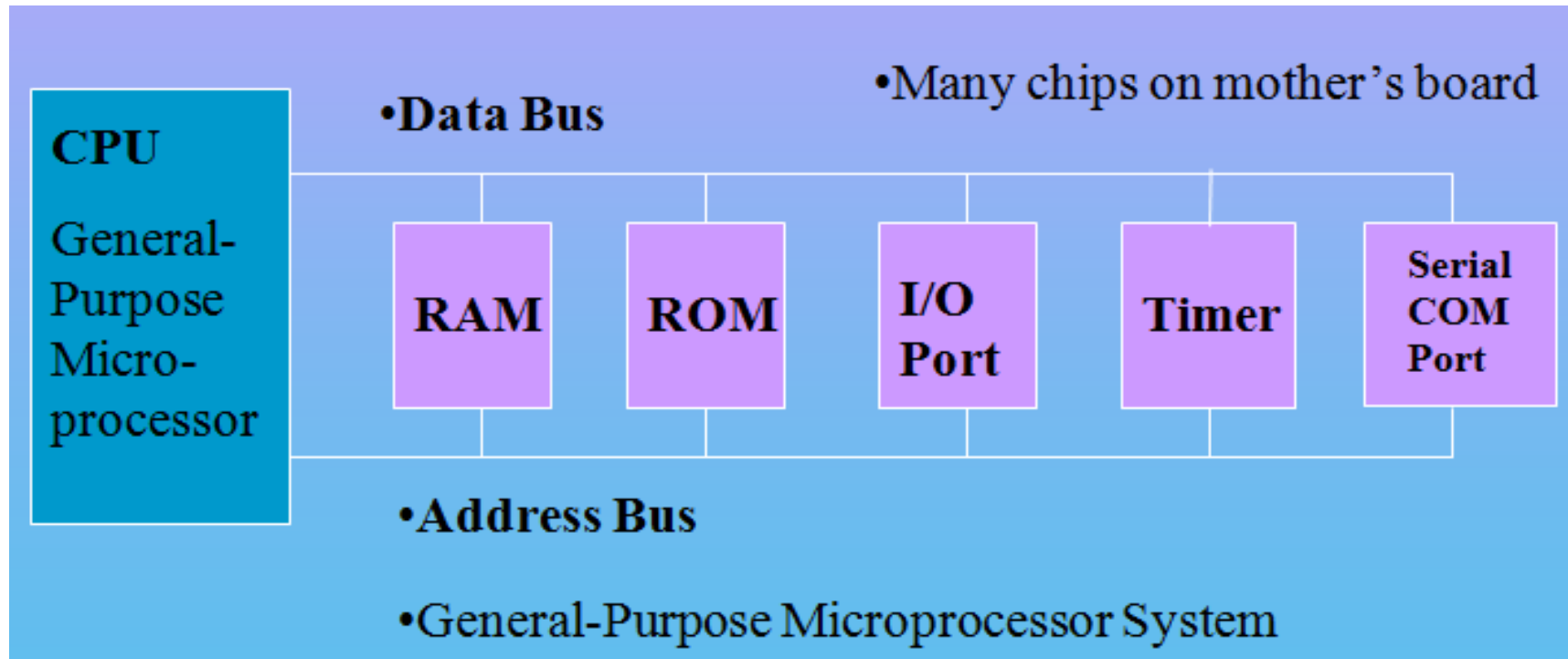
Microprocessor(cont'd.)

▶ Stack Pointer (SP)

- It is a small register that stores the address of the last program/instruction request in a stack.

Microprocessor(cont'd.)

- CPU for Computers
- No RAM, ROM, I/O on CPU chip itself
- Example: Intel's x86, Motorola's 680x0



Microcontroller

Micro



Very Small

Controller



A mechanism
that controls
the operation of
a machine

Microcontroller(cont'd.)

- A microcontroller is a highly integrated single chip, which consists of on chip CPU (Central Processing Unit), RAM (Random Access Memory), EPROM/PROM/ROM (Erasable Programmable Read Only Memory), I/O (input/output) – serial and parallel ports, timers, interrupt controller.
- For example, Intel 8051 is 8-bit microcontroller and Intel 8096 is 16-bit microcontroller.

Microcontroller(cont'd.)

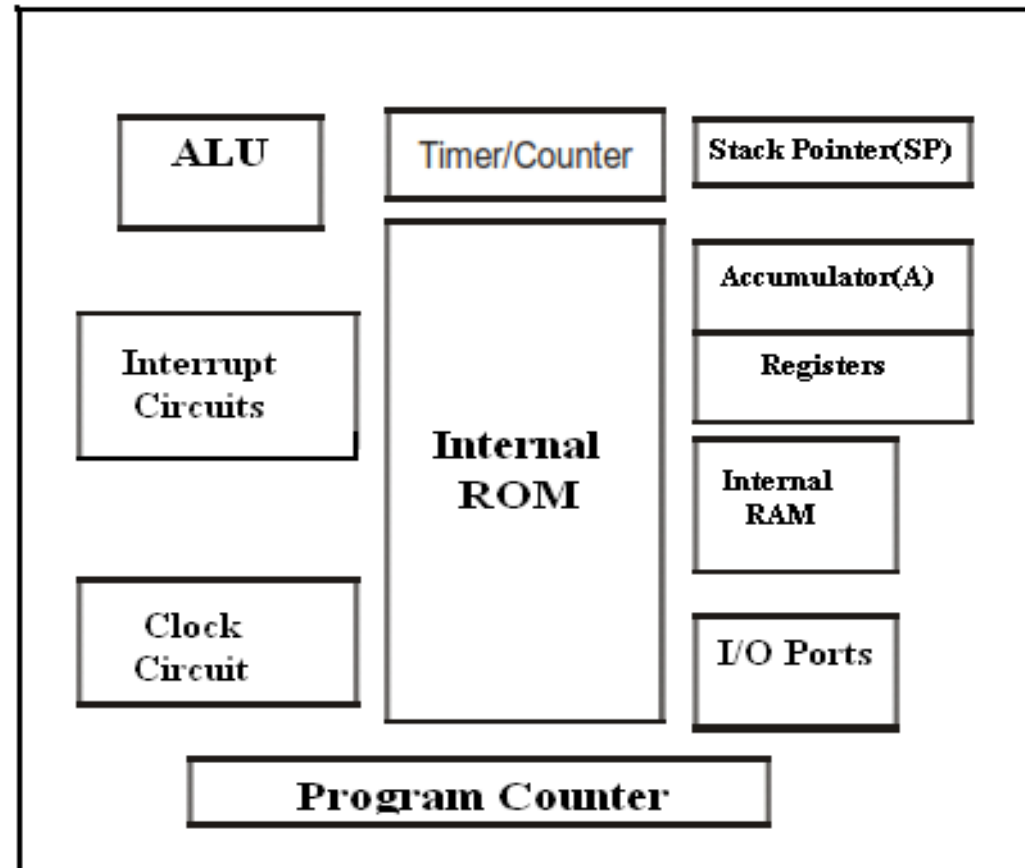


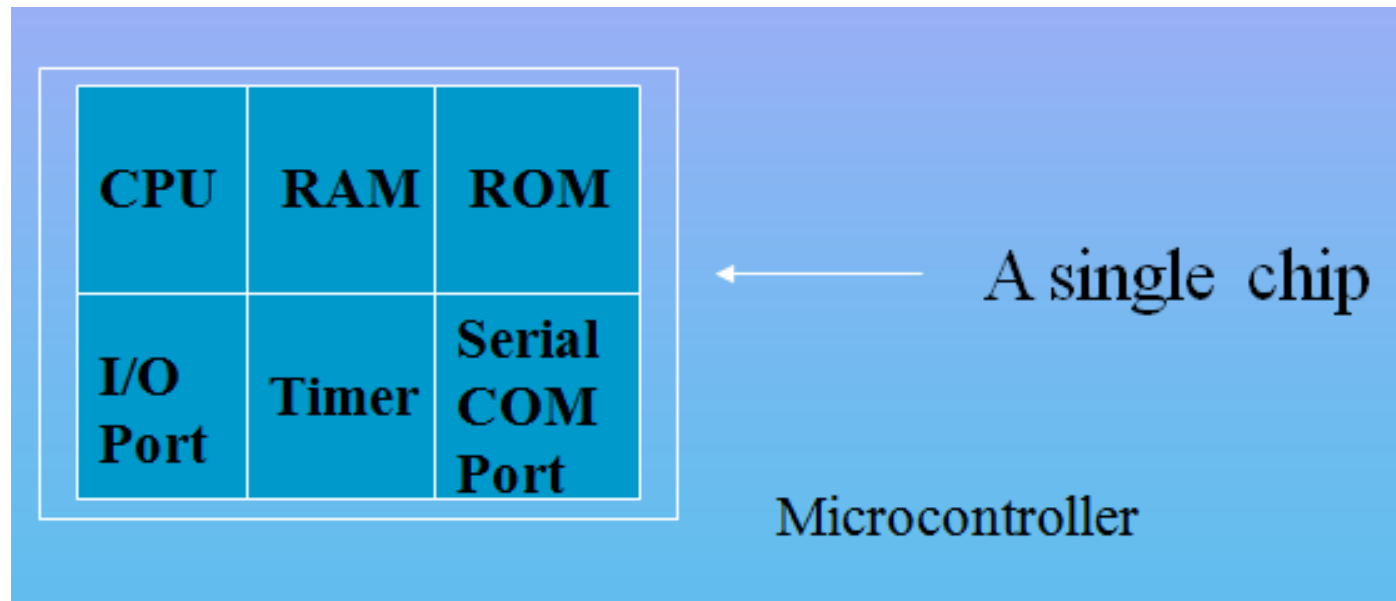
Fig. Block Diagram of a Microcontroller

Applications of microcontrollers

- ▶ Cell phone
- ▶ Pager
- ▶ Watch
- ▶ Calculator
- ▶ video games
- ▶ Alarm clock
- ▶ Air conditioner
- ▶ TV remote
- ▶ Microwave oven
- ▶ Washing machines
- ▶ An electronic smart weight display system
- ▶ Robotic system
- ▶ An anti-lock braking system monitor

Microcontroller(cont'd.)

- A smaller computer
- On-chip RAM, ROM, I/O ports...
- Example: Motorola's 6811, Intel's 8051, Zilog's Z8 and PIC



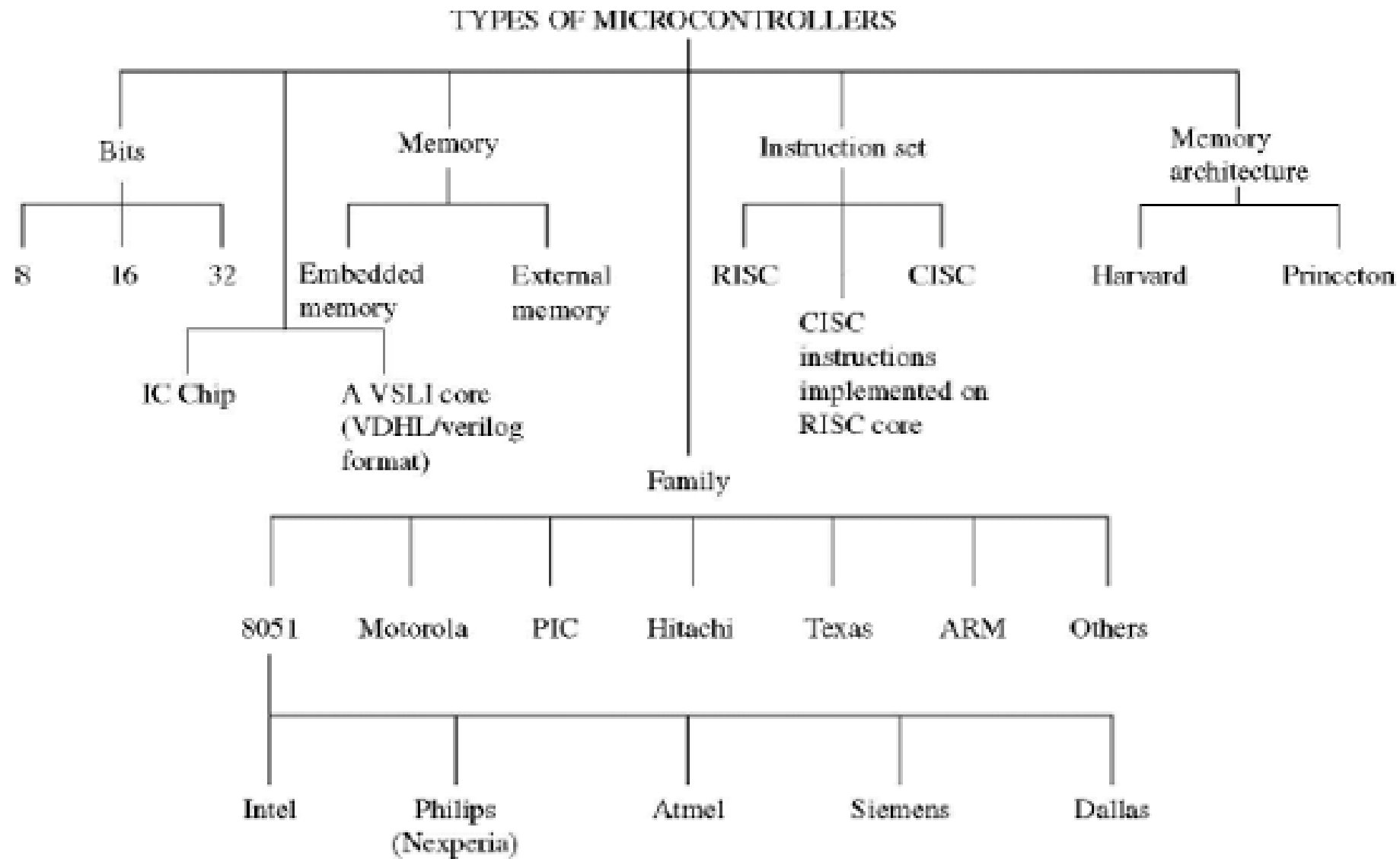
Microprocessor vs. Microcontroller

Microprocessor

- ▶ CPU is stand-alone, RAM, ROM, I/O, timer are separate
- ▶ designer can decide on the amount of ROM, RAM and I/O ports.
- ▶ expensive
- ▶ general-purpose
- ▶ High processing power
- ▶ High power consumption
- ▶ Instruction sets focus on processing-intensive operations
- ▶ Typically 16/32/64 – bit
- ▶ Typically deep pipeline (5–20 stages)

Microcontroller

- CPU, RAM, ROM, I/O and timer are all on a single chip
- fixed amount of on-chip ROM, RAM, I/O ports
- for applications in which cost, power and space are critical
- single-purpose (control-oriented)
- Low processing power
- Low power consumption
- Bit-level operations
- Instruction sets focus on control and bit-level operations
- Typically 8/16 bit
- Typically single-cycle/two-stage pipeline



Introduction to RISC and CISC

▶ RISC:

It stands for **Reduced Instruction Set Computer**. It is a type of microprocessor architecture that uses a small set of instructions of uniform length. These are simple instructions that are generally executed in one clock cycle. RISC chips are relatively simple to design and inexpensive. The setback of this design is that the computer has to repeatedly perform simple operations to execute a larger program having a large number of processing operations.

Examples: SPARC, POWER PC etc.

Introduction to RISC and CISC (cont'd.)

▶ CISC:

It stands for **Complex Instruction Set Computer**. These processors offer the users, hundreds of instructions of variable sizes. CISC architecture includes a complete set of special-purpose circuits that carry out these instructions at a very high speed. These instructions interact with memory by using complex addressing modes. CISC processors reduce the program size and hence lesser number of memory cycles are required to execute the programs. This increases the overall speed of execution.

Examples: Intel architecture, AMD

RISC	CISC
Focus on software.	Focus on hardware.
Uses only Hardwired control unit.	Uses both hardwired and microprogrammed control unit.
Transistors are used for more registers	Transistors are used for storing complex Instructions
Fixed sized instructions	Variable sized instructions
Can perform only Register to Register Arithmetic operations	Can perform REG to REG or REG to MEM or MEM to MEM
Requires more number of registers	Requires less number of registers
Code size is large	Code size is small

An instruction executed in a single clock cycle	Instruction takes more than one clock cycle
An instruction fit in one word.	Instructions are larger than the size of one word
Simple and limited addressing modes.	Complex and more addressing modes.
RISC is Reduced Instruction Cycle.	CISC is Complex Instruction Cycle.
The number of instructions are less as compared to CISC.	The number of instructions are more as compared to RISC.
It consumes the low power.	It consumes more/high power.
RISC is highly pipelined.	CISC is less pipelined.
RISC required more RAM.	CISC required less RAM.
Here, Addressing modes are less.	Here, Addressing modes are more.

Thank You